MIL-T-43819A
30 March 1984
SUPERSEDING
MIL-T-43819(GL)
27 October 1972

MILITARY SPECIFICATION

TRAY, MESS, 5-COMPARTMENT, PAPER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a disposable, paper, 5-compartment mess tray (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7350

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DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

5-11-1175 - Tray Mess, 5-Compartment, Paper

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 170-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The trays shall be the manufacturer's standard commercial product, except for any changes necessary to meet document requirements.

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3.3 Certification.

3.3.1 Federal Food, Drug and Cosmetic Act Certification. Prior to the approval of the first article, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the molded pulp for the trays and the anti-oxidants or other substances incorporated in the molded pulp he proposes to supply meet the requirements of Federal Food, Drug, and Cosmetic Act as Amended: Title 21, CFR 176.170. In addition, the exact composition for the disposable trays shall be furnished to the contracting officer for evaluation to determine safe incineration. Acceptable evidence of meeting these requirements shall be a certified test report from a recognized independent testing laboratory acceptable to the Department of the Army Surgeon General.

3.4 Materials (see 6.5). Pulp for the trays shall be produced from chemical or mechanical pulp. Additives or after-treatments may be used provided the materials, processees, and finished tray comply with all provisions of this document.

3.5 Design and construction. The compartmented tray shall conform to Drawing 5-11-1175. The compartmented tray shall be molded construction and the trays shall be nestable.

3.5.1 Rigidity. Trays shall not show a deflection greater than 3/8-inch when tested as specified in 4.5.1.

3.5.2 Water resistance. Trays shall not show total penetration of water, ply separation or delamination more than 1/4 inch in any dimension, or any blister more than 1¼ inch in diameter, when in contact with water for 30 minutes and at a water temperature between 160°F and 162°F when tested as specified in 4.5.2. Total penetration is defined as the wetting and staining of the bond paper under the tray.

3.5.3 Odor and taste. Trays shall not have any objectionable odor and shall not impart any objectionable taste to water when tested as specified in 4.5.3.

3.5.4 Grease resistance. The tray shall be grease resistant and shall not show evidence of total penetration by oil when tested as specified in 4.5.4. Total penetration is defined as the wetting and staining of the bond paper under the tray.

3.6 Finish. The tray shall have a hot die dried pressed finish. Color shall be white or natural pulp unless otherwise specified. The tray shall be free from dust, dirt, grease, oil, foreign matter, and slime spots.

3.7 Marking for identification. The trays shall be marked with the manufacturer's name, trade name, or trademark readily identifiable with the manufacturer.

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3.8 Workmanship. All paper stock for the trays shall be protected from contamination during and after processing and shall be clean and free from holes, tears, cuts, cracks, scuffs, or buckled areas, delamination, blister, distortaion, malformation or warpage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and 4.4.3 and tested for the characteristics specified in table II. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.1.1 Certification. A certificate of compliance will be acceptable as evidence of conformance to the requirements of 3.3.1.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of one tray. The sample unit shall be one completely fabricated tray. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. Visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not as specified	X	
	Not color specified		X
	Rough edges	X	
	Not free of dust, dirt, grease, oil, or foreign matter imbedded in molded pulp	X	
Construction and workmanship	Not design specified	X	
	Number of compartments not as specified	X	
	Trays do not nest	X	
	Not clean and free from holes, tears, cuts, cracks, scuffed or buckled areas	X	
	Any delamination, blister, distortion, malformation, or warpage	X	
Marking	Missing, not legible, incomplete, or not as specified		X

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified a defect. The lot size shall be expressed in units of one box. The sample unit shall be one completely fabricated tray. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units shall be 2.5.

4.4.4 End item testing. The end item shall be tested for the characteristics specified in table II. The lot size shall be expressed in units of trays. The sample unit shall be one tray. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Rigidity	3.5.1	4.5.1
Water resistance	3.5.2	4.5.2
Odor and taste	3.5.3	4.5.3
Grease resistance	3.5.4	4.5.4

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4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with section 5 requirements. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully packaged. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE III. Packaging defects

Examination	Defect
Marking (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing or damaged.
Contents	Number per container is more or less than required.
Workmanship	Incomplete closure, inadequate stapling, loose strapping or improper taping Bulged or distorted container

4.4.5.1 Examination for palletization. An examination shall be made to determine that palletization complies with section 5 requirements. Defects shall be as indicated in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Palletization defects

Examine	Defect
Finished dimension	Length, width, or height, exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified

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TABLE IV. Palletization defects (cont'd)

Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection

4.5.1 Rigidity test

4.5.1.1 Apparatus. The tray to be tested shall be held in a paper plate rigidity tester with extra long hold down bar (see 6.4) as illustrated on figure 1.

4.5.1.2 Procedure. Place the tray with the eating surface upward on the support wedge with stops set for a 9-3/4 inch diameter plate (4-7/8 inch mark) and the 8-1/2 inch side with the two small compartments against the stops. Lower the hold-down bar or clamp over the top of the tray so that it makes contact with the tray rim. Apply a weight of 150 grams at the outer edge of the rim and measure the deflection of the tray caused by the weight, to the nearest 1/32 inch to determine compliance with 3.5.1. Inability of the tray to meet the specified requirements for rigidity shall constitute failure of this test.

4.5.2 Water resistance test. Distilled water at a temperature between 160°F and 162°F shall be poured onto the tray surfaces to a depth approximately equal to one-half the depth of the tray. The tray shall rest on a flat sheet of double face corrugated board or other suitable paper support, approximately 1/4 inch thick, with a sheet of clean white bond paper between the tray and supporting surface. After 30 minutes the water in the test tray shall be drained into a flask and retained for use in the odor and taste test. The test tray shall then be examined for evidence of delamination. Separation of plies extending more than 1/4 inch in any direction or formation of blisters greater than 1/4 inch in diameter, shall be considered delamination. The bond paper and under-side of the tray shall be examined for visible evidence of water leakage to determine compliance with 3.5.2. Slight dampness of the paper due to condensation shall not be construed as a failure. Inability to meet the requirements for water resistance specified shall constitute failure of this test.

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4.5.3 Odor and taste test. The distilled water used in water resistance test (4.5.2) shall be collected and allowed to stand in a stoppered flask or bottle for 2 hours. The odor and taste of the sample water shall be compared with that of freshly distilled water at the same temperature to determine compliance with 3.5.3. Inability to meet the requirements for odor and taste specified shall constitute failure of this test.

4.5.4 Grease resistance test. Place a mixture of 3.0 milliliters of corn oil, of the grade used for cooking, and 10 grams of clean, dry sand, of such a particle size that all of it will pass through as No. 20 U.S. Standard sieve and be retained by a No. 30 U.S. Standard sieve, in a glass beaker of approximately 30 milliliter capacity and heat in an oven to 160°F ± 2°F. Remove the hot mixture from the oven and swirl the beaker contents to assure good mixing of the oil and sand. Cover the beaker with the tray to be tested, using the eating surface of the tray as the cover. Invert the tray and beaker and place the tray on a hard, flat surface with a sheet of clean white bond paper directly under the tray, with the beaker in an inverted position over the sand and oil mixture. Allow the mixture to stand, as specified above, for 30 minutes. At the end of this test period, examine the bond paper and the bottom of the tray for any evidence of staining, to determine compliance with 3.5.4. Inability to meet the requirements for grease resistance specified shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be Commercial.

5.1.1 Commercial. Trays shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Five hundred trays preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Five hundred trays, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW), or type SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

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5.2.3 Commercial packing. Trays, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2) trays packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. Compartmented paper trays are intended for one-time use in military feeding systems and all systems using the Tray Pack Ration.

6.2 Ordering data. Acquisition document should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of the applicable level of packing (see 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of three trays. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Rigidity test. The apparatus refered to in 4.5.1.1 may be obtained from the Single Service Institute 250 Park Ave., New York, NY 10017.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.4).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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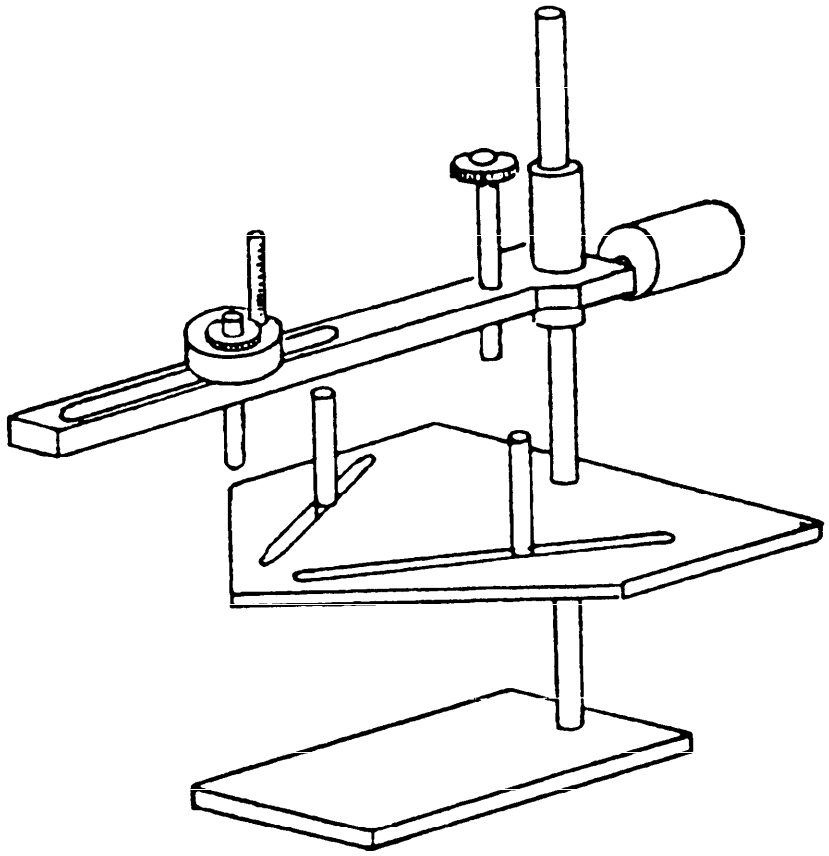


FIGURE 1 - PAPER PLATE RIGIDITY TESTER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-43819A	Tray, Mess, 5-Compartment, Paper

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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